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Personalized Directional Human Effective Mass Estimation for Online Collision-Force Assessment and Speed Scaling of Collaborative Robots

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Abstract: Collaborative robots using power and force limiting adjust their speed to keep the estimated collision force with a human worker within biomechanical limits. In online frameworks, the robot-side effective mass can be evaluated for the current configuration and contact direction, whereas the human-side term is fixed at a body-region value or removed by a clamped-contact assumption. This paper presents an online method supplying the human-side term from the worker's current state. A Skinned Multi-Person Linear (SMPL) model is reconstructed as a personalized 19-segment multibody system, and a constrained-dynamics projection incorporates foot support and restricts disallowed motion directions at joint range-of-motion (ROM) boundaries selected by the collision direction. The resulting directional human effective mass enters the existing reduced-mass, collision-force, allowable-velocity, and speed-scaling calculation without changing its structure. Within one segment, effective mass varied about 2.7-fold with collision direction and 2.46-fold with knee ROM activation. At decision-boundary speeds defined by fixed-mass force limits and along the effective-mass-maximizing directions of crouched-head and raised-arm postures, the directional estimate increased the estimated force by 41.57% and 57.78%. In an online Robot Operating System 2 (ROS 2) implementation on a collaborative manipulator, the estimated force differed from the fixed-mass baseline by 18.06% on average over the run's head-approach interval, with a mean software-side assessment-to-command latency of 26.84 ms. The Jacobian-based velocity mapping and a chest comparison with a body mass index (BMI)-based reference effective mass showed mean relative errors of 3.31% and 5.59%. The force differences are model-output comparisons, not measured collision-force validation.

Keywords: Collaborative robot, constrained dynamics, human effective mass, physical human–robot interaction, power and force limiting, speed scaling.

1. INTRODUCTION

Collaborative robots operate in close proximity to human workers, which increases the possibility of unintended physical contact and makes collision safety assessment a fundamental requirement for human–robot collaboration [1]. ISO 10218-2:2025 specifies safety requirements for collaborative industrial robot applications, including power and force limiting (PFL), and incorporates the PFL provisions for physical contact originally given in ISO/TS 15066 [2, 3].

In PFL-based collision assessment, the force generated during a potential collision depends on the relative velocity, contact stiffness, and reduced mass of the robot–human system [2, 4]. The reduced mass is determined jointly by the robot effective mass and the human effective mass at the contact site, so that the allowable robot velocity depends directly on the human-side term [5]. On the human side, standard PFL-based assessment commonly uses body-region-specific fixed masses [3], and online PFL frameworks either fix this term at a single worst-case or representative body-region value [6, 7] or remove it through a clamped-contact assumption [8]. These treatments do not reflect body shape, posture, collision direction, support conditions, or joint range-of-motion (ROM) boundaries, while perception-driven robot motion broadens the range of approach directions and human configurations. Treating every contact as clamped, which corresponds to an infinite human effective mass, can make the assessment of unconstrained contacts overly conservative and limit robot efficiency, and injury studies on unconstrained collisions have called for better models of the effective masses of human body parts [9].

Parametric human body models provide subject-specific geometry, pose, and body-segment properties that can support dynamic human modeling [10–12]. On the

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robot side, configuration- and direction-dependent effective mass is already evaluated online in PFL-based monitoring and collision reaction [7, 8], and the tracked human pose is used for separation distances, body-part limits, and contact-type classification [6, 13]. What remains open is to derive the human-side effective mass from the current state of a specific worker, including support conditions and joint ROM boundaries, and to supply it online to PFL-based collision-force estimation and speed scaling.

This paper addresses this problem with an online method that supplies a personalized directional human effective mass to PFL-based speed scaling. The main contributions are as follows. First, the human-side term of the PFL reduced mass is updated online per state within a Robot Operating System 2 (ROS 2) speed-scaling pipeline on a collaborative manipulator, without changing the existing reduced-mass, allowable-velocity, and speed-scaling structure. Second, this term is computed from a personalized Skinned Multi-Person Linear (SMPL)-based 19-segment multibody model at the segment center of mass and along the collision direction, in the same operational-space sense as the robot-side effective mass. Third, a direction-dependent ROM-active constraint selection rule is formulated within a constrained-dynamics projection, in which a joint limit is activated according to the sign of the joint acceleration induced by a unit contact force; a standard property of the constrained inverse inertia (Proposition 1) ensures that each activated limit cannot lower the directional effective mass. The method is evaluated through model-level consistency analyses, a body mass index (BMI)-based chest reference comparison, controlled fixed-mass comparisons, and an online ROS 2 implementation.

2. RELATED WORK

2.1. Online PFL-Based Collision-Force Assessment

Haddadin et al. scaled the commanded velocity from the reflected robot mass along the contact normal, assuming a stationary human [14]. Online PFL frameworks evaluate the robot-side term as the simplified ISO/TS 15066 mass [6], a direction-dependent effective mass [7, 8], or link kinetic energy [13], whereas the human-side term, where it enters the force computation, is a fixed body-region value or is removed by a clamped-contact assumption (Table 1). For constrained (pinching) contacts, a contact-force model predicts the clamping force from the force–deformation response at the contact and the robot's stopping behavior, and thereby a safe velocity [15].

2.2. Effective Mass and Human Body Modeling

Khatib's operational-space formulation projects joint-space inertia through a task-space Jacobian [16,17]; the effective mass along a direction is the inverse of directional mobility and varies with configuration, evaluation point, and contact direction. Robot-side studies have examined the error of the simplified ISO/TS 15066 mass model [18], direction-averaged reflected mass [19], and safety maps combining robot dynamics with injury data [20]. Here, effective mass denotes the directionally projected operational-space inertia; reflected mass and mean reflected mass denote the direction-dependent and direction-averaged robot-side quantities of [18–20].

On the human side, contact models for physical human–robot interaction commonly assign lumped mass, stiffness, and damping to body regions, limiting their treatment of worker-specific variability [21]. Parametric body models supply individual shape and pose [10], per-body-part mass and inertia [12], and subject-specific segment parameters [11]; shape and pose can be estimated online [22], and anatomical landmarks can be mapped to the mesh [23]. Support conditions and joint ROM boundaries alter the human inertial response at the contact point, which is distinct from classifying a contact as free or clamped [13]; the present formulation addresses the former, and Table 1 positions it against the fixed or clamped-contact human-side treatments of standard practice and online PFL frameworks [3, 6–8].

3. PERSONALIZED DIRECTIONAL HUMAN EFFECTIVE MASS FOR PFL SPEED SCALING

3.1. Framework Overview

The framework (Fig. 1) has a human module that reconstructs the SMPL-based 19-segment model and applies structural, foot-support, and ROM constraints to obtain the directional human effective mass, a robot module that computes the directional effective mass of the corresponding link, and an assessment module that forms the reduced mass, estimated collision force, and allowable approach velocity for each retained link–segment pair, the most restrictive pair setting the speed-scaling factor. The human-side effective mass is the only quantity added; the rest keeps its standard PFL form.

3.2. Personalized 19-Segment Human Body Model

SMPL shape parameters determine body geometry and joint locations, and pose parameters the relative configuration of body segments [10]. The 24-joint structure is reconstructed as a 19-segment rigid-body model by aggregating distal hand and foot joints into terminal segments and reorganizing the spine into trunk segments; the segmented mesh supplies the regions used for segment mass properties. We adapt the Phys-SMPL representation [12], which augments SMPL geometry with per-body-part mass and inertia, from 24 body parts to the 19 segments. Segment volume, center of mass, and inertia tensor follow from tetrahedral accumulation over each segment mesh [24], and segment mass from a reference body-mass distribution with shape-dependent volume scaling and tissue-density factors [12, 25]. The kinematic reconstruction is

Table 1. Positioning of this work relative to prior effective-mass and online PFL studies

	Effective-mass formulation			*Online PFL application*		
Study	Robot side	Human side	Human-side dependence	Human effective-mass update	Human-state input	Robot command
Mansfeld [20], Kirschner [18]	Direction-dependent		Not formulated		–	–
Steinecker [19]	Direction-averaged		Not formulated		–	–
Svarny [6]	Non-direction-dependent (ISO/TS 15066)	Body-region constant (chest; ISO/TS 15066)	None (worst case)	Offline, fixed	Skeleton keypoints (separation distance, body-part limits)	Speed reduction, stop
Lucci [8]	Direction-dependent	Infinite (clamped contact)	None (worst case)	Offline, fixed	Point cloud with body-part labels (separation distance, body-part limits)	Speed scaling
Rustler [7]	Direction-dependent (link motion)	Body-region constant (whole arm, 5.6 kg; ISO/TS 15066)	None (single region)	Offline, fixed	–	Stop/continue
Thumm [13]	Energy-based (link kinetic energy)		Not formulated		Pose (reachable occupancy, contact type)	Speed scaling
This work	Direction-dependent	Directional effective mass (constrained multibody)	Body shape, posture, contact direction, constraints	Online, per state	SMPL shape and pose (human effective mass)	Speed scaling

Note: A human-side inertial term counts as formulated only when it enters the reduced-mass, collision-force, or collision-energy computation of that study's proposed method; injury data, force and pressure limits, and energy thresholds are not human inertial terms. "Infinite" denotes a term removed by a clamped-contact assumption ($1/m_{\mathrm{h}}^{\mathrm{eff}} = 0$). Human-state input is online sensing used by the proposed method, its purpose in parentheses.

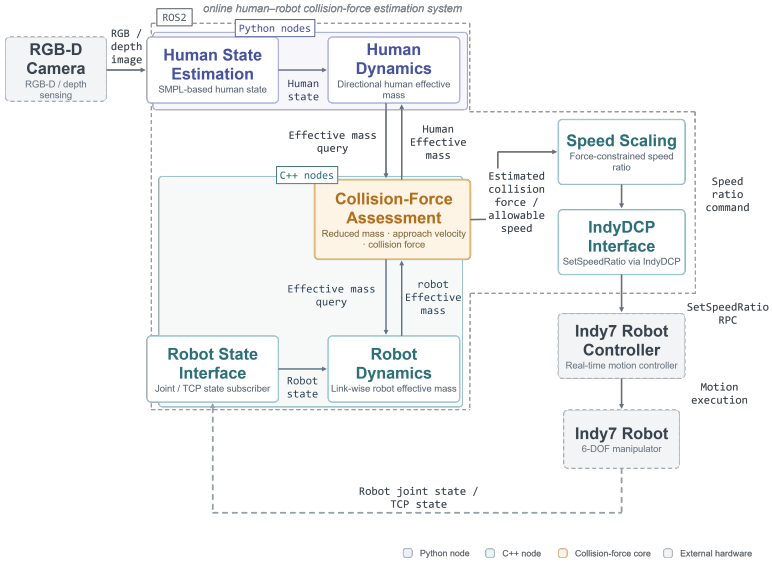


Fig. 1. ROS 2-based online collision-force assessment and speed-scaling architecture.

verified in Section 4.1; the segment inertial parameters are adopted from the Phys-SMPL distribution and the density factors cited above and are not independently identified here.

3.3. Human Mass Matrix and Contact Jacobian

The model has $\mathbf{q}_{\mathrm{h}} \in \mathbb{R}^{60}$: a pelvis floating base with six degrees of freedom (DOFs) and 54 internal rotational coordinates, with velocity $\dot{\mathbf{q}}_{\mathrm{h}}$. The segment center-of-mass (CoM) Jacobians $\mathbf{J}_{v,i}, \mathbf{J}_{\omega,i} \in \mathbb{R}^{3\times 60}$ follow from the standard recursion over the reconstructed tree, and the human mass matrix $\mathbf{M}_{\mathrm{h}}(\mathbf{q}_{\mathrm{h}})$ assembles the translational and rotational inertia of the 19 segments through them [26]. For an evaluation point $\mathbf{p}_k$ on segment k, the point Jacobian $\mathbf{J}_{p,k} \in \mathbb{R}^{3\times 60}$ gives $\dot{\mathbf{p}}_k = \mathbf{J}_{p,k}(\mathbf{q}_{\mathrm{h}})\dot{\mathbf{q}}_{\mathrm{h}}$.

3.4. Constrained-Dynamics Projection

The admissible motion space is restricted by the model structure, bilateral foot support, and joint ROM boundaries. These are incorporated by sequential projection of the inverse mass matrix, which gives the local acceleration response permitted by the active constraints; it does not simulate post-impact motion or estimate constraint forces. An active constraint set is written at the velocity level as $\mathbf{A}(\mathbf{q}_{\mathrm{h}})\dot{\mathbf{q}}_{\mathrm{h}} = \mathbf{0}$. Eliminating its reaction force with the standard force projector [26] leaves the force-to-acceleration map, which for a general mobility matrix $\mathbf{B}$ defines

$$\mathcal{P}(\mathbf{B},\mathbf{A}) = \mathbf{B} - \mathbf{B}\mathbf{A}^{\mathsf{T}}\left(\mathbf{A}\mathbf{B}\mathbf{A}^{\mathsf{T}}\right)^{-1}\mathbf{A}\mathbf{B}, \tag{1}$$

applied sequentially, starting from $\mathbf{B} = \mathbf{M}_{\mathrm{h}}^{-1}$, to the structural, foot-support, and ROM-active constraints with a regularized solver when active rows are nearly dependent. Structural constraints remove coordinates that are not independent DOFs; the knee, for example, keeps only flexion–extension. Foot support constrains the translational velocities of both foot-segment CoMs through the stacked Jacobian $\mathbf{J}_{\mathrm{f}} \in \mathbb{R}^{6\times 60}$, three rows per foot, leaving foot rotation free. With the structural selection matrix $\mathbf{A}_{\mathrm{s}}$, $\mathbf{M}_{\mathrm{s}}^{-1} = \mathcal{P}(\mathbf{M}_{\mathrm{h}}^{-1}, \mathbf{A}_{\mathrm{s}})$ and $\mathbf{M}_{\mathrm{f}}^{-1} = \mathcal{P}(\mathbf{M}_{\mathrm{s}}^{-1}, \mathbf{J}_{\mathrm{f}})$.

ROM-active constraints are then selected in parent-to-child order from $\mathbf{M}_{\mathrm{cur}}^{-1} \leftarrow \mathbf{M}_{\mathrm{f}}^{-1}$. For a point on segment k and unit direction $\mathbf{u}$, the acceleration induced by a unit directional force under the constraints activated so far is $\ddot{\mathbf{q}}_{\mathrm{dir}} = \mathbf{M}_{\mathrm{cur}}^{-1}\mathbf{J}_{p,k}^{\mathsf{T}}\mathbf{u}$. For a rotational DOF d with admissible range $[q_d^{\min}, q_d^{\max}]$ and boundary tolerance $\varepsilon_{\mathrm{ROM}}$, a ROM row is activated when

$$s_d = \operatorname{sgn}(\ddot{q}_{\mathrm{dir},d}) \neq 0 \text{ and } s_d\left(q_d - q_d^{\lim}(s_d)\right) \geq -\varepsilon_{\mathrm{ROM}}, \tag{2}$$

with $q_d^{\lim}(+1) = q_d^{\max}$ and $q_d^{\lim}(-1) = q_d^{\min}$: the force drives the DOF outward within the tolerance; a DOF with $\ddot{q}_{\mathrm{dir},d} = 0$ is left inactive. The evaluation used $\varepsilon_{\mathrm{ROM}} = 5^\circ$ and the model's joint-specific ROM limits. This induced acceleration omits the velocity-dependent terms and the constraint bias, so the activation test depends on posture and collision direction but not on the current joint velocities; in the posture of Fig. 4 the knee is fully extended, recorded as 0° [27], which is the model's extension limit, so activation there does not depend on $\varepsilon_{\mathrm{ROM}}$. Activated rows are projected out before downstream joints are evaluated, giving $\mathbf{A}_{\mathrm{rom}}(\mathbf{u})$ and $\mathbf{M}_{\mathrm{c}}^{-1}(\mathbf{q}_{\mathrm{h}}, \mathbf{u}) = \mathcal{P}(\mathbf{M}_{\mathrm{f}}^{-1}, \mathbf{A}_{\mathrm{rom}}(\mathbf{u}))$.

If no boundary is activated, $\mathbf{M}_{\mathrm{c}}^{-1} = \mathbf{M}_{\mathrm{f}}^{-1}$ (Algorithm 1). Because $\ddot{\mathbf{q}}_{\mathrm{dir}}$ is re-evaluated after each projection, the active set can depend on the evaluation order; the implementation uses the fixed parent-to-child order of the kinematic chain from the pelvis to segment k, and only the joints of that chain are candidates. The selection is thus a single parent-to-child pass that approximates the active set of the unilateral joint limits: rows activated earlier are not rechecked after later projections, DOFs left inactive at their turn are not revisited, and joints outside the chain are not candidates.

Proposition 1: Let $\mathbf{B} \succeq \mathbf{0}$ be a mobility matrix, let $\mathbf{A}$ satisfy $\mathbf{A}\mathbf{B}\mathbf{A}^{\mathsf{T}} \succ \mathbf{0}$, and let $\mathbf{u}$ be a unit collision direction at contact point k. If $\mathbf{u}^{\mathsf{T}}\mathbf{J}_{p,k}\,\mathcal{P}(\mathbf{B},\mathbf{A})\,\mathbf{J}_{p,k}^{\mathsf{T}}\mathbf{u} > 0$, then

$$\left[\mathbf{u}^{\mathsf{T}}\mathbf{J}_{p,k}\,\mathcal{P}(\mathbf{B},\mathbf{A})\,\mathbf{J}_{p,k}^{\mathsf{T}}\mathbf{u}\right]^{-1} \geq \left[\mathbf{u}^{\mathsf{T}}\mathbf{J}_{p,k}\,\mathbf{B}\,\mathbf{J}_{p,k}^{\mathsf{T}}\mathbf{u}\right]^{-1},$$

that is, adding an ideal constraint does not decrease the directional effective mass.

Proof: By (1), $\mathbf{B} - \mathcal{P}(\mathbf{B},\mathbf{A}) = \mathbf{B}\mathbf{A}^{\mathsf{T}}(\mathbf{A}\mathbf{B}\mathbf{A}^{\mathsf{T}})^{-1}\mathbf{A}\mathbf{B} \succeq \mathbf{0}$. With $\mathbf{w} = \mathbf{J}_{p,k}^{\mathsf{T}}\mathbf{u}$, this gives $0 < \mathbf{w}^{\mathsf{T}}\mathcal{P}(\mathbf{B},\mathbf{A})\mathbf{w} \leq \mathbf{w}^{\mathsf{T}}\mathbf{B}\mathbf{w}$, and taking reciprocals reverses the inequality. Applying the result to the structural and foot-support projections and to each projection in Algorithm 1 extends it to all three constraint sets. □

Algorithm 1 Direction-Dependent ROM-Active Constraint Selection

Require: $\mathbf{q}_{\mathrm{h}}$, unit collision direction $\mathbf{u}$, contact Jacobian $\mathbf{J}_{p,k}$, $\mathbf{M}_{\mathrm{f}}^{-1}$, joint ROM limits, $\varepsilon_{\mathrm{ROM}}$
Ensure: $\mathbf{A}_{\mathrm{rom}}(\mathbf{u})$, $\mathbf{M}_{\mathrm{c}}^{-1}(\mathbf{q}_{\mathrm{h}}, \mathbf{u})$
$\mathbf{M}_{\mathrm{cur}}^{-1} \leftarrow \mathbf{M}_{\mathrm{f}}^{-1}$, $\mathcal{I}_{\mathrm{rom}} \leftarrow \emptyset$
for each joint j in parent-to-child order **do**
  $\ddot{\mathbf{q}}_{\mathrm{dir}} \leftarrow \mathbf{M}_{\mathrm{cur}}^{-1}\mathbf{J}_{p,k}^{\mathsf{T}}\mathbf{u}$
  $\mathcal{I}_{\mathrm{new},j} \leftarrow$ DOFs of j satisfying (2)
  if $\mathcal{I}_{\mathrm{new},j} \neq \emptyset$ **then**
    build $\mathbf{A}_{\mathrm{new},j}$ from the selection rows of $\mathcal{I}_{\mathrm{new},j}$
    $\mathbf{M}_{\mathrm{cur}}^{-1} \leftarrow \mathcal{P}(\mathbf{M}_{\mathrm{cur}}^{-1}, \mathbf{A}_{\mathrm{new},j})$
    $\mathcal{I}_{\mathrm{rom}} \leftarrow \mathcal{I}_{\mathrm{rom}} \cup \mathcal{I}_{\mathrm{new},j}$
  end if
end for
build $\mathbf{A}_{\mathrm{rom}}(\mathbf{u})$ from the rows of $\mathcal{I}_{\mathrm{rom}}$; $\mathbf{M}_{\mathrm{c}}^{-1} \leftarrow \mathbf{M}_{\mathrm{cur}}^{-1}$
return $\mathbf{A}_{\mathrm{rom}}(\mathbf{u})$, $\mathbf{M}_{\mathrm{c}}^{-1}(\mathbf{q}_{\mathrm{h}}, \mathbf{u})$

By Proposition 1, the support and ROM constraints move the estimate toward the constrained end, and adding a further ideal constraint to the selected set cannot decrease the directional effective mass; if bracing is idealized as locking additional joints, the reported value is therefore a lower bound for that braced state. The proposition concerns ideal structural, support, and ROM constraints only; passive joint stiffness, muscle activation, and tissue response are not modeled and lie outside its scope.

3.5. Personalized Directional Human Effective Mass

Projecting $\mathbf{M}_{\mathrm{c}}^{-1}$ through the point Jacobian gives the Cartesian inverse inertia at $\mathbf{p}_k$, and, for a unit collision direction $\mathbf{u} \in \mathbb{R}^3$, the scalar human effective mass is the reciprocal of the directional mobility,

$$\mathbf{\Lambda}_{\mathrm{h},k}^{-1}(\mathbf{q}_{\mathrm{h}}, \mathbf{u}) = \mathbf{J}_{p,k}\mathbf{M}_{\mathrm{c}}^{-1}(\mathbf{q}_{\mathrm{h}}, \mathbf{u})\mathbf{J}_{p,k}^{\mathsf{T}}, \quad m_{\mathrm{h},k}^{\mathrm{eff}} = \left[\mathbf{u}^{\mathsf{T}}\mathbf{\Lambda}_{\mathrm{h},k}^{-1}\mathbf{u}\right]^{-1}. \tag{3}$$

This is the operational-space definition used on the robot side [16, 17], evaluated on the constrained personalized human model. For visualization, with posture and active set held fixed, the eigenvalues of $\mathbf{\Lambda}_{\mathrm{h},k}$ give the principal effective-mass ellipsoid [17].

3.6. Collision-Force Assessment and Speed Scaling

To limit online queries, the nearest robot link is found for each human segment and the pairs are sorted by distance; the top-K pairs form the candidate set $\mathcal{C}$, with $K = 3$ here. Let $\mathbf{p}_{\mathrm{r},l}$ and $\mathbf{p}_{\mathrm{h},k}$ be the centers of mass of robot link l and human segment k, used alike for distance, direction, approach speed, and effective mass. Assessment runs only when the minimum pairwise distance $D_{\min} = \min_{(l,k)\in\mathcal{C}} \|\mathbf{p}_{\mathrm{h},k} - \mathbf{p}_{\mathrm{r},l}\|$ lies within an activation radius $R_{\mathrm{act}} = 1.300$ m, a computational gating threshold; pairs outside it impose no speed constraint, so its value does not alter the assessment of pairs inside it. The unit collision direction $\mathbf{u}_{lk} = (\mathbf{p}_{\mathrm{h},k} - \mathbf{p}_{\mathrm{r},l})/\|\mathbf{p}_{\mathrm{h},k} - \mathbf{p}_{\mathrm{r},l}\|$ points from the link point to the segment point, and the approach

speed is the closing component of the relative velocity along it, $v_{\mathrm{app},lk} = \max\big(0, \mathbf{u}_{lk}^{\mathsf{T}}(\dot{\mathbf{p}}_{\mathrm{r},l} - \dot{\mathbf{p}}_{\mathrm{h},k})\big)$.

The robot effective mass $m_{\mathrm{r},l}^{\mathrm{eff}}$ along $\mathbf{u}_{lk}$ is computed as in (3) with the translational Jacobian $\mathbf{J}_{\mathrm{r},l} \in \mathbb{R}^{3\times 6}$ at $\mathbf{p}_{\mathrm{r},l}$ and the joint-space inertia $\mathbf{M}_{\mathrm{r,aug}} = \mathbf{M}_{\mathrm{r}}(\mathbf{q}_{\mathrm{r}}) + \mathbf{M}_{\mathrm{g}}$ of the six-axis Indy7, augmented by the diagonal gear-reflected actuator inertia $\mathbf{M}_{\mathrm{g}}$ [4, 17]; the reduced mass of the pair is

$$\mu_{lk} = \left[\frac{1}{m_{\mathrm{r},l}^{\mathrm{eff}}} + \frac{1}{m_{\mathrm{h},k}^{\mathrm{eff}}}\right]^{-1}, \tag{4}$$

evaluated along the same direction for the current states. With κ_k and F_k^{lim} the body-region contact stiffness and permissible force limit of segment k (the skull-and-forehead region for the head segment), and with the fixed human masses used only in the baseline, all following ISO/TS 15066 as incorporated into the PFL provisions of ISO 10218-2:2025 [2, 3], the lumped linear-compliance impact model [2, 4] gives

$$F_{\mathrm{est},lk} = v_{\mathrm{app},lk}\sqrt{\mu_{lk}\,\kappa_k}, \qquad v_{\mathrm{allow},lk} = \frac{F_k^{\mathrm{lim}}}{\sqrt{\mu_{lk}\,\kappa_k}}. \tag{5}$$

A pre-emptive factor $0 < \eta \le 1$ starts deceleration early, giving, for $v_{\mathrm{app},lk} > 0$ (with $s_{lk}^* = 1$ otherwise), the pairwise and raw scales

$$s_{lk}^* = \min\left(1, \frac{\eta\, v_{\mathrm{allow},lk}}{v_{\mathrm{app},lk}}\right), \qquad s_{\mathrm{raw}} = \min_{(l,k)\in\mathcal{C}} s_{lk}^*, \tag{6}$$

and an asymmetric first-order low-pass filter smooths the applied command, $s_{t+1} = s_t + \alpha_t(s_{\mathrm{raw}} - s_t)$ with $\alpha_t = \alpha_{\mathrm{dec}}$ if $s_{\mathrm{raw}} < s_t$ and α_{acc} otherwise, $\alpha_{\mathrm{dec}} > \alpha_{\mathrm{acc}}$, so deceleration is rapid and recovery gradual. The demonstration used $\eta = 0.40$, $\alpha_{\mathrm{dec}} = 0.28$, and $\alpha_{\mathrm{acc}} = 0.10$, empirical system-specific values that act on the speed command but not on the force assessment. The applied factor s_{t+1} is sent to the controller.

4. EVALUATION

The evaluation comprises model-level checks and online integration tests rather than measured collision-force validation.

4.1. Kinematic Consistency Analysis

The Jacobian velocity mappings were compared with finite-difference velocities from corresponding SMPL motions in AMASS [28], over 8,548 sequences and 3,214,306 frames from nine subsets spanning diverse body shapes and motions. Because adjacent frames are temporally correlated, this is a large-scale consistency check, not an independent statistical validation. The Jacobian-based estimates $\hat{\mathbf{x}}_i$, obtained from forward-differenced joint coordinates, were compared with a forward finite difference of the segment CoM positions computed from the posed SMPL mesh in each frame and, for rotation, with $\boldsymbol{\omega}_i^{\mathrm{ref}}(t) = \mathrm{Log}^{\vee}(\mathbf{R}_i(t+\Delta t)\mathbf{R}_i(t)^{\mathsf{T}})/\Delta t$, through the relative error $e_i^x(t) = \|\hat{\mathbf{x}}_i(t) - \mathbf{x}_i^{\mathrm{ref}}(t)\|_2 / \|\mathbf{x}_i^{\mathrm{ref}}(t)\|_2$, $\mathbf{x}_i \in \{\mathbf{v}_i, \boldsymbol{\omega}_i\}$.

Over the nine subsets (ACCAD [29], BioMotionLab NTroje [30], CMU [31], EKUT and KIT [32], Eyes Japan [33], MPI Limits [34], TCD handMocap [35], TotalCapture [36]), the overall linear and angular relative errors were 3.31% (1.79–4.56% across subsets) and 0.69% (0.21–1.02%), with Euclidean errors of 0.0084 m/s and 0.0062 rad/s. The reconstructed tree and its Jacobians therefore map both velocities consistently, supporting the 19-segment kinematic reconstruction; this check does not assess the segment inertial parameters.

4.2. Reference-Based Comparison of Chest Effective Mass

The chest effective mass was compared with the BMI-based thoracic reference parameter of Dong et al. [37] to examine body-shape-dependent scaling consistency, not subject-specific impact response. Following the cadaveric experiments that calibrated the reference, the evaluation point is the fourth costal interspace at the sternum [38], located on the SMPL mesh by the landmark mapping of [23] (one vertex index for all samples). Matching the reference's lower-body support and unconstrained back, the model was seated with the thigh-segment support points constrained in translation and the chest effective mass evaluated along the anterior-to-posterior direction; this approximates, but does not reproduce, all cadaver-test boundary conditions. With the body mass index (BMI) the body weight W over the squared height H, both computed from the SMPL shape of each sample, the reference effective mass is $m_{\mathrm{h,ref}}^{\mathrm{chest}} = \mathrm{BMI}\,W/87$.

The reference is an empirical lumped-thorax equivalent mass calibrated against cadaveric force–deflection data [37], whereas the proposed value is an instantaneous constrained operational-space effective mass, so the comparison is one of magnitude and scaling trend, not of accuracy; because the reference is a deterministic function of body mass and height, a linear regression on body mass alone already reaches $R^2 = 0.863$. The check is therefore whether the constrained operational-space computation, which uses no thoracic calibration, lands in the same range (estimates of 12.24–31.25 kg against reference values of 11.99–33.84 kg) and follows the same anthropometric scaling; it does. The 272 unique AMASS/SMPL body-shape sets (subsets of Section 4.1, duplicates removed) cover the source pool's anthropometric variation and do not represent an industrial-worker population; over them the mean absolute, root-mean-square, and mean relative errors (MAE, RMSE, MRE) were 1.13 kg, 1.47 kg, and 5.59%, with $R^2 = 0.818$ about the line $y = x$ (Fig. 2), and at higher reference masses the estimates fell below the reference.

4.3. Directional and Constraint-Dependent Effective Mass

At a head evaluation point, the directional inertial distribution is visualized by the ellipsoid of Section 3.5, whose

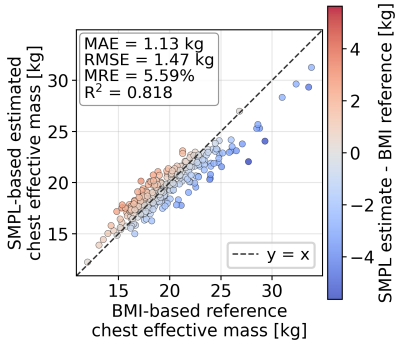


Fig. 2. Chest effective mass over 272 body-shape samples, reference versus estimate. The diagonal is $y = x$; color is the estimate–reference difference.

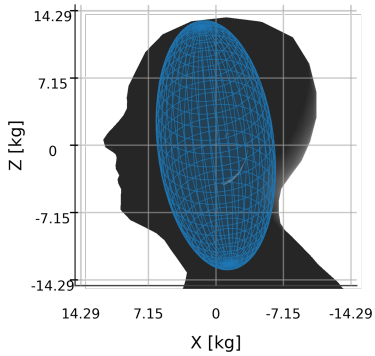


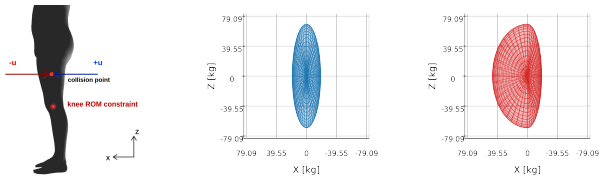
Fig. 3. X–Z view of the mass-scaled principal effective-mass ellipsoid at the head evaluation point. The semi-axis lengths are the eigenvalues of the translational operational-space inertia in kilograms.

longer semi-axes indicate lower mobility. For the posture in Fig. 3, the principal effective masses ranged from 6.17 to 16.51 kg, a ratio of about 2.7, which demonstrates directional variation within a single segment.

The knee case was evaluated at a thigh point under opposite directions $\mathbf{u}$ and $-\mathbf{u}$ (Fig. 4(a)); a knee DOF becomes active when the directional force would drive the joint beyond its admissible range. Under the same posture, point, direction, and foot support, knee ROM activation increased the scalar directional thigh effective mass from 19.01 to 46.84 kg, a factor of about 2.46 (Fig. 4(b) and (c)). The activated constraint removes the blocked motion component, consistent with Proposition 1, and yields an equal or more restrictive allowable velocity. Under the same posture, point, and $-\mathbf{u}$ direction, the effective mass rose from 37.46 kg without foot support to 46.84 kg with bilateral support, again consistent with Proposition 1 as model-output sensitivity to the prescribed support condition.

4.4. Controlled Comparison Against a Fixed Human Mass

For each retained pair of the online demonstration, the collision force was computed with either the proposed or the corresponding body-region fixed human



(a) Direction and knee ROM constraint. (b) Ellipsoid without ROM constraint. (c) Ellipsoid with ROM constraint.

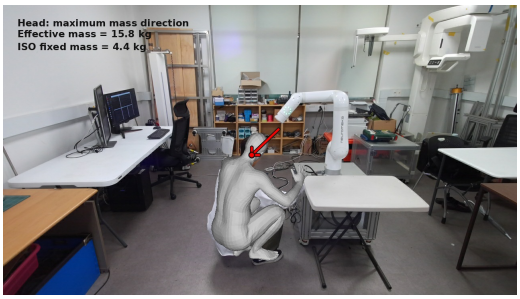
Fig. 4. Effect of knee ROM activation on the thigh effective mass. Panels (b) and (c) use the same posture, evaluation point, collision direction, and foot-support condition as (a); in (c), the knee DOF is active for only one of the two opposite directions, so the surface is not centrally symmetric.

Table 2. Controlled fixed-mass discrepancy for the head (skull and forehead) and upper arm (upper arms and elbow joints); body-region values from ISO/TS 15066:2016 Tables A.2 and A.3 [3] as incorporated into ISO 10218-2:2025 [2]. All values are model outputs, not measured collision forces.

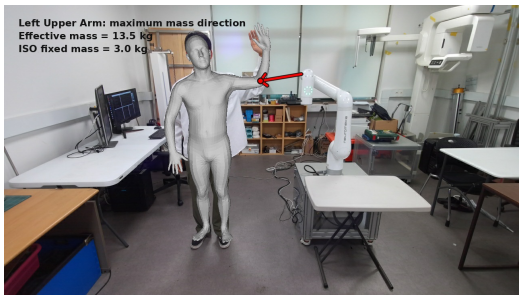
	Head	Upper arm
Shared conditions (identical in both calculations)		
Robot effective mass [kg] (representative value)	10.0	10.0
Contact stiffness [N/mm]	150	30
Evaluation speed [m/s]	0.192	1.140
Force limit [N]	130 (quasi-static)	300 (transient)
Human effective mass (the only quantity varied)		
Fixed [kg]	4.4	3.0
Proposed [kg]	**15.8**	**13.5**
Model output		
F_{est}, fixed mass [N]	130.0	300.0
F_{est}, proposed [N]	**184.0**	**473.3**
Force discrepancy [%]	**41.57**	**57.78**

mass, all other inputs identical; the baseline used the body-region parameters of ISO/TS 15066 as incorporated into ISO 10218-2:2025 [2, 3], and pressure limits were not evaluated. For frames with positive approach velocity, the relative discrepancy $\Delta F_{\text{rel}}(t) = |F_{\text{est,dir}}(t) - F_{\text{est,ISO}}(t)|/F_{\text{est,ISO}}(t) \times 100$ was evaluated, with $F_{\text{est,dir}}$ and $F_{\text{est,ISO}}$ the estimates from the proposed and fixed human masses; non-approaching frames were excluded because both estimates are zero. Over the head-approach interval of the demonstration (3.9–10.7 s, Fig. 7) the discrepancy was 18.06% on average and 23.60% at maximum, with a standard deviation of 4.16% and a median of 15.37%, over 167 frames of one run lasting 16.2 s; in 1.2% of these frames the two calculations reached different decisions on whether additional speed reduction was required. The discrepancy is attributable to the human-side mass representation because all other inputs were unchanged.

Two controlled cases were also evaluated, a crouched head posture and a raised-arm upper-arm posture, each with the collision direction chosen to maximize the directional human effective mass (Fig. 5; directionally severe conditions that do not represent the contact-direction distribution of the online run). For comparison, the direction-



(a) Head, crouched posture.



(b) Upper arm, raised-arm posture.

Fig. 5. Controlled maximum-effective-mass cases for the head (crouched) and upper arm (raised arm); arrows indicate the maximizing directions.

averaged values at the same points are 9.6 kg (head) and 9.3 kg (upper arm), and the online run's most restrictive segment ranged from 7.73 to 8.65 kg, so the maxima in Table 2 combine state dependence with the choice of direction. The robot effective mass was fixed at 10.0 kg, a representative value chosen for the comparison, and only the human-side mass varied; the evaluation speed was the baseline allowable value, at which the fixed-mass calculation equals the force limit. The head used the quasi-static skull-and-forehead limit, as no transient value is specified for that region and head contact is not an admissible PFL contact case; the head serves here and in Section 4.5 to illustrate the speed-scaling response during approach, not a permitted contact. The upper arm used the transient limit for a moving, non-clamping contact.

Since the evaluation speeds follow from the fixed-mass force limits, these are decision-boundary conditions: at the same speed, the human-side representation changes whether additional speed reduction is required. The sign of the change follows whether the directional value exceeds the tabulated mass, and the robot effective mass sets its size through the harmonic combination: the head and upper-arm values exceed the tabulated masses of 4.4 and 3.0 kg and raise the estimate, whereas the thigh values of Section 4.3 (19.01–46.84 kg) fall below the 75 kg tabulated mass, which equals the full body weight [3], and lower it by 3.4–13.8% at a robot effective mass of 10.0 kg. Over the online robot effective-mass range of the selected link, 3.97–20.66 kg, the discrepancies in Table 2 would range from 23.3% to 57.1% for the head and from 34.0% to 76.5% for the upper arm.

4.5. Online Implementation and Timing

The framework was implemented as the asynchronous ROS 2 pipeline of Section 3.1 and run in a shared workspace with an Indy7 manipulator and a single worker; bilateral foot support was assumed rather than sensed. The human state was obtained from an Azure Kinect DK in narrow-field-of-view unbinned depth mode, mounted 2.435 m from the robot base at a height of 1.55 m, tilted 20.8° downward, with a fixed camera-to-base transform. The shape parameters were estimated with CLIFF [39], averaged over the first ten frames, and then held fixed;

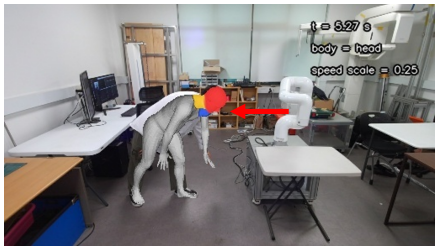


Fig. 6. Online run of Section 4.5 at the head approach, with the tracked model overlaid (applied speed scale 0.25).

the pose was regressed in each frame from the Azure Kinect body-tracking joints [22]; shape and pose were published at 30 Hz. For the participant, the model gave a height of 1.80 m and a total mass of 73.1 kg. The depth coverage on a horizontal plane at torso height (1.20 m) contains 85.2% of the activation region, and all frames in which the assessment was active lay inside it. Tracking loss is not explicitly handled (the dynamics node republishes the last pose with zero velocity, and an unanswered query batch drives the speed scale to its minimum after 150 ms); none occurred in the reported run. Assessment ran while the worker was inside the activation region of radius $R_{\mathrm{act}} = 1.300$ m, with the top-three candidates tracked continuously (Fig. 6). The online demonstration is a single run of 16.2 s (631 assessment snapshots); the force statistics of Section 4.4 are evaluated over its 3.9–10.7 s head-approach interval shown in Fig. 7. Timing statistics in Table 3 were recorded in a separate session with the same pipeline configuration.

Table 3. Software-side latency of the assessment path per query batch; P95 denotes the 95th percentile, and maxima are observed peaks, not timing guarantees.

Stage	Mean [ms]	P95 [ms]	Max [ms]	N
Assessment and command latency				
Human effective-mass query	19.46	25.87	43.63	2064
Robot effective-mass query	2.64	3.87	40.16	2064
Candidate-pair assessment	25.18	30.09	49.20	2064
Assessment-to-command	26.84	32.12	53.72	2064

The human sensing and human dynamics nodes are driven by 30 Hz timers. In Table 3, the assessment-

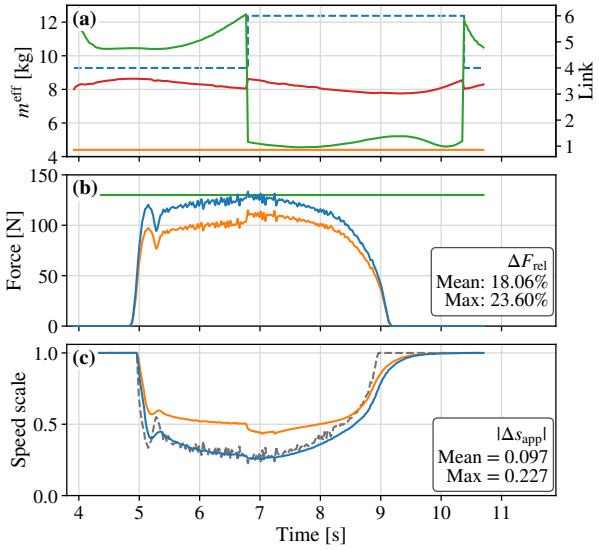


Fig. 7. Online response for the most restrictive candidate pair. (a) Directional human (red), robot (green), and fixed head (orange) effective masses with the selected link (blue dashed, right axis). (b) Force estimates from the proposed (blue) and fixed (orange) masses against the permissible-force reference (green), annotated with ΔF_{rel} for $v_{\text{app}} > 0$. (c) Raw (grey dashed) and applied (blue) speed-scaling factors, the latter also recomputed offline with the fixed mass (orange).

to-command path had a mean of 26.84 ms and a P95 of 32.12 ms, measured from query issuance to speed-command completion, and so excludes the RGB-D (color and depth) sensing stage before it and the physical robot response after it. Because the two queries are asynchronous, these latencies are wall-clock values, not sums of the query latencies. The human state used by a command is older than this path: the dynamics-state stamp to command completion was 42.9 ms on average and 79.7 ms at maximum, the sensing stage adds at least one 30 Hz frame period (47.1% of sensing callbacks exceeded the 33.3 ms timer period), and the capture-to-command age was not measured directly.

In Fig. 7, the most restrictive candidate changed with the evolving geometry, while the directional human effective mass stayed above the fixed head mass over the shown interval, and the applied scale fell to 0.25 as the robot approached the worker's head and recovered to 1.00 as it moved away. The offline fixed-mass trajectory in Fig. 7(c) is an open-loop recomputation along the recorded run in which only the human-side mass was replaced; all other inputs matched the online run, and the applied speed-scaling trajectory then differed from the online one by 0.097 in mean absolute value and 0.227 at maximum.

5. DISCUSSION

Because the human-side term enters the collision-force and allowable-velocity relations, the variation reported in Section 4 propagates to the speed-scaling command. The contribution lies in formulating and integrating this state-dependent representation online, yielding an auditable term that PFL-based monitors, speed-scaling laws, and safety-constrained controllers or planners [40,41], including perception-driven pipelines that resolve the worker only to body-part labels [42], can consume.

Because the reduced mass combines the two effective masses harmonically, a change in the human term matters less when the robot effective mass is the smaller, so the values should not be generalized independently of robot configuration and contact direction. Only the force component of PFL is modeled, with body-region stiffness and force limits taken from the standard and held fixed, so that supplying a different human-side mass isolates the inertial term. At a common approach speed, the force discrepancy depends only on the two reduced masses and not on the force limit or contact stiffness, so limit values refined by human-subject studies [43,44] would shift the evaluation speeds in Table 2 but not its discrepancies. Foot support was prescribed rather than estimated (Section 4.3 shows a change from 37.46 kg to 46.84 kg for one posture and direction), the online experiment was a single-subject prototype, and tracking loss is not explicitly handled. Collision experiments with instrumented force measurement are planned as the next step, to relate the directional estimate to measured collision forces across subjects, postures, and contact directions; online support-state estimation will be added in the same study.

6. CONCLUSION

This paper presented an online method that supplies a personalized directional human effective mass to PFL-based collision-force assessment and speed scaling. A personalized SMPL-based 19-segment multibody model and a constrained-dynamics projection account for body shape, posture, collision direction, foot support, and joint ROM boundaries, and the direction-dependent ROM-active selection rule activates joint limits within this projection, each activated limit being unable to lower the directional effective mass. The resulting term updates the human side of the existing reduced-mass, allowable-velocity, and speed-scaling chain online without changing its structure. Model-level checks, a controlled fixed-mass comparison, and an online ROS 2 implementation on a collaborative manipulator show that the term can be supplied online within the software-side latency reported in Section 4.5; relating it to measured collision forces is the planned next step.

DECLARATIONS

Conflict of Interest. Heonseop Shin is employed by Safetics Inc., Seoul, Republic of Korea, which is active in collaborative-robot safety assessment; this affiliation is disclosed as a potential competing interest. The other authors declare no competing financial or personal interests.

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Authors' Contributions. H. Kim conceived the study, developed the framework, and led the implementation, experiments, and analysis. J. Lee contributed to the conceptualization and research direction; J. Choi to the Jacobian formulation and kinematic validation; H. Shin to extending the directional effective-mass formulation to a constrained human multibody framework. S. Rhim supervised the study. All authors approved the final manuscript.

Ethics and Consent. Ethics approval was not required because the study involved no medical intervention, invasive procedure, or biological sample. The analysis used public datasets, and the online demonstration was performed by Hyeonuk Kim, an author, who voluntarily participated and gave informed consent to the use and publication of the images in which he appears.

Data Availability. The motion sequences are from the public AMASS dataset (`https://amass.is.tue.mpg.de`); SMPL is available from its original distributor under the applicable license. Logs, processed data, and implementation details are available from the corresponding author on reasonable request; the code is not publicly released because it includes system-specific robot-interface components.

DECLARATION OF AI TOOL USE

The authors used generative AI tools (OpenAI ChatGPT and Anthropic Claude) for language editing, structural refinement, reference verification, and formatting of the manuscript. The authors reviewed and verified all content and take full responsibility for it.

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